Assessment Proforma 2025-26

Key Information

Module Code	CMT223
Module Title	Internet of Things
Module Leader	Charith Perera
Module Moderator	Xianfang Sun
Assessment Title	Group IoT Development Project
Assessment Number	1 of 1
Assessment Weighting	100%
Assessment Limits	Poster A1 Size Demo Video 5 mins MAX CodeCast Video 20 mins MAX

This assessment's due date and time is specified on Learning Central. To view it click on the assessment's link, which will open *'Details & Information'* section for this assessment. Alternatively, go to *'Calendar'* section of a module, and select *'Due Dates'* option. You can change the view to daily or monthly, as well as use arrows to navigate the calendar.

The COMSC Assessment Calendar for all assessments is also available on Learning Central: by going to '**COMSC-SCHOOL'** organisation -> **'Assessment & Feedback 2025 - 26'** folder -> **'COMSC Assessment Calendar 2025 - 26'** page.

Learning Outcomes

The learning outcomes for this assessment are as follows:

1. Critically dissect and analyse existing IoT systems in terms of the design choices.
2. Analyse and apply common design decisions, their trade-offs, and consequences.
3. Identify and formulate a problem in the area of IoT.
4. Design, develop and deploy an IoT application and justify their design choices.
5. Create reproducible projects with well-documented codebases.
6. Disseminate project outcomes to a broader audience through digital means.
7. Awareness of relevant professional, ethical, legal and social issues that arise in the implementation of existing and future computer systems

Submission Instructions

All assessment files should be submitted in '*Assessment & Feedback'* folder on Learning Central.

Before you submit your assessment, you **must complete and submit** *'Assessment Coversheet'* form. Further instructions are given in this module's 'Assessment & Feedback' folder.

Your submission for this assessment should contain the following file(s):

Description		**Type**	**Name**
Coversheet	Compulsory	One PDF (.pdf) file	Coversheet.pdf
Poster	Compulsory	One PDF (.pdf) file	GroupID_Poster.pdf
Demo	Compulsory	One MP4 (.mp4) file	GroupID_Demo.mp4
CodeCast	Compulsory	One MP4 (.mp4) file	GroupID_CodeCast.mp4
SourceCode	Compulsory	One ZIP (.zip) file	GroupID_SourceCode.zip

If you are unable to submit your work, email your submission to the module leader and comsc-submissions@cardiff.ac.uk. If the technical issue is with Cardiff University systems, also report it to the University IT Support: email it-support@cardiff.ac.uk, tel. +44 (0)29 2251 1111.

- No poster = incomplete submission (typical rules for late submission applies).
- No Demo Video = incomplete submission (typical rules for late submission applies).
- No Code Cast Video = incomplete submission (typical rules for late submission applies).
- No Git Source Code Repo = incomplete submission (typical rules for late submission applies).

Recommendations from University Digital Education Support Team: The university recommends you follow this guideline: https://xerte.cardiff.ac.uk/play_18610#page4 Under the heading Submitting your Assessment, there is a section containing guidance for students on making Panopto assignment submissions. There's an instructional video there with clear instructions on what to do.

Should the assessment team encounter issues accessing any submitted files, such as corruption, we will deem that submission incomplete. It falls upon the students to provide all necessary files in advance and verify their accessibility. We recommend re-downloading each file after submission to confirm they are intact and readable. Avoid last-minute

uploads which could result in unforeseen issues and leave no time for rectification. The Learning Central submission system is configured so that students can submit required files unlimited times, and the assessment team will look at only the last submission. Therefore, students are encouraged to submit their work early and test that everything is correctly submitted.

Assessment Description

Task: Students are required to develop an Internet of Things (IoT) system using a provided IoT hardware kit. This kit includes microcontrollers, single-board computers, sensors, actuators, and access to IoT cloud platforms. This assessment aims to allow students to demonstrate their ability to design, implement, and present a functional IoT system of their choice.

The following steps are merely suggestions; students are welcome to adopt any suitable methodology to achieve the above task.

Brainstorming Ideas (Step 1): Brainstorming ideas for an IoT system begins with identifying potential problems or inefficiencies that could be addressed using IoT technology. Engage in creative thinking sessions to generate a variety of ideas, considering different applications and fields where IoT could make a significant impact. You will also need to consider the iot kit half and how to best utilise it to develop the potential idea. You will need to think about iot system within the constraints of the iot kit, which has limited number of sensors and actuators.

Defining the Objective (Step 2): Once you have selected a promising idea, clearly define the objective of your IoT system. Determine whether your goal is monitoring, control, automation, or data collection, etc.. A well-defined objective will guide the development process and ensure the system meets its intended purpose.

Developing Requirements (Step 3): With a clear objective in mind, develop a comprehensive list of specific requirements to ensure your IoT system effectively achieves its intended purpose. Begin by identifying strong user requirements and user needs. Understanding user needs is crucial for defining the functionality and usability of your system. You are also encouraged to identify both functional and non-functional requirements of your iot system and make design decisions accordingly.

Feature Design (Step 4): Breaking down the requirements into distinct features is crucial in developing a comprehensive IoT system. Students can systematically address each aspect of the project by deconstructing the system's objectives and requirements into manageable, functional units. This approach simplifies the development process and ensures that each feature is thoroughly designed, implemented, and tested. Viewing the system as an incorporation of various features allows for a modular design, where

individual features can be developed and improved independently while maintaining a cohesive integration within the overall system.

Guidance for Developing Features

- In this coursework, the concept of a "feature" is important. In this coursework, a feature is defined as a unit of functionality or capability. For example, you can consider login as a feature of your IoT system.

- **Balanced Approach**: While some aspects of a feature may require more detailed discussion than others, ensure that you cover all the aspects listed above.

- **Quality Over Quantity**: Examiners will focus on the depth and difficulty of implementation of each feature rather than the sheer number of features. Avoid developing many simple features; instead, aim for a few well-implemented, complex features.

- **Creativity and Complexity**: Some features may be easy to develop using existing frameworks and APIs. While these are acceptable, you may need to develop more such features to score higher marks. Strive to include features that showcase a range of skills and demonstrate your ability to handle varying complexities.

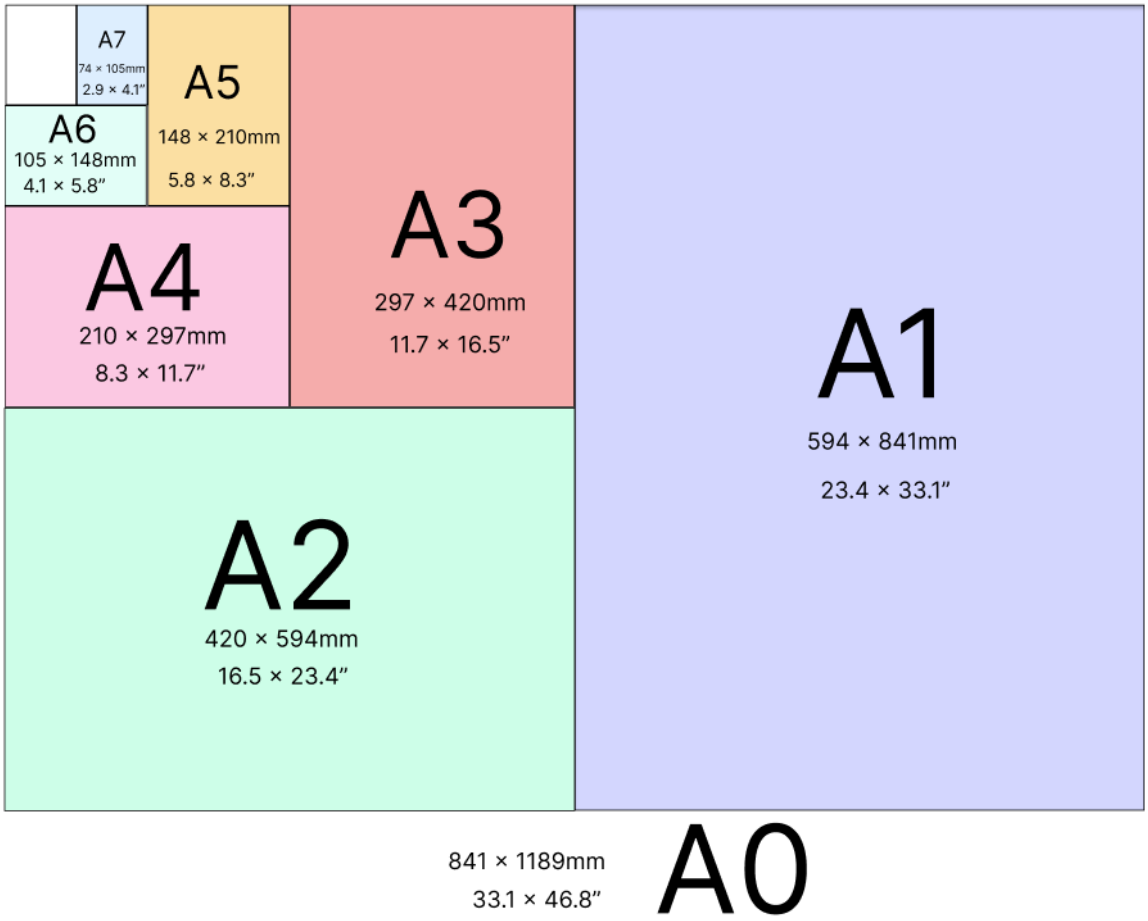
- **Skill Demonstration**: Choose features that align with your existing skill set and challenge you to learn new skills. This will help you demonstrate competencies across all seven aspects taught in the module.

- Your IoT system should demonstrate a range of skills by developing appealing and relevant features that address the problem your system is intended to solve. Focus on creating a well-rounded project that highlights your ability to handle tasks of different complexities while applying the knowledge gained throughout the course.

- **Uniquely Numbering:** Each feature must have a unique number and a short name (e.g., (1) Login). This number and the short name should be used across all submission material to refer to a given feature.

Poster (one file in PDF format)

- In this coursework, the term **'poster'** is used interchangeably with **'written submission'**.
- Please ensure that the poster is created exclusively in A1 dimensions. Consult the size guide below for precise measurements.



Software Choice: You are free to use any software you prefer to design your poster. However, Microsoft PowerPoint is the recommended default option due to its user-friendly features and wide availability.

- **Required Information**: Your poster must include the following elements:
  1. **Project Title**: Clearly state the title of your project.
  2. **Domain**: Identify the domain to which your IoT system belongs (e.g., healthcare, agriculture, security).
  3. **System Description**: Provide a detailed description of the IoT system you developed. Explain its purpose and functionality.

4. **Value Proposition**: Describe the value proposition of your IoT system. What benefits or improvements does it offer to its users?
5. **Architecture/Data Flow Diagram**: Include a diagram that visualises and explains your IoT application's architecture and data flow. This diagram should be labelled:
    - All components of the system
    - Data types collected
    - Sensors and actuators used
    - Network protocols and communication technologies employed
    - Data analytics processes (e.g., how data is processed and utilised)
    - Security and privacy techniques implemented
6. **Critical Discussion of Each Feature:** You are required to critically discuss each feature of your project from the following aspects:
    - **Functionality**: What does the feature do? How does it contribute to the overall system?
    - **Implementation**: How was the feature developed? What technologies and methods were used? Implementation details
    - **Complexity**: What is the level of difficulty involved in developing this feature?
    - **Quality**: How well is the feature implemented? Consider robustness, efficiency, and user experience.
    - **Relevance**: How relevant is the feature to the problem your system addresses?
    - **Creativity**: How innovative is the feature?
    - **Application of Course Concepts**: How does the feature apply what you've learned in the course?
7. **Practical Application of Course Concepts for Each Feature:** When discussing each feature of your IoT project, address the following aspects to demonstrate how you applied course concepts in practice:
    - **Architecture**: How was this specific feature designed and structured within the overall system?
    - **Sensing and Actuation**: What methods were used for sensing and actuating related to this feature?

- **Networking and Communications**: How did you handle networking and communication for this feature between components?
- **Data Management and Analytics**: How did you manage and analyse data for this feature within your system?
- **Privacy and Security**: What measures did you implement to ensure privacy and security for this feature?
- **Human Factors and Interaction**: How did your design of this feature consider human factors and user interaction?
- **Design Strategies and Prototyping**: What strategies did you use in this feature's design process and prototyping?

- **Text and Readability:** There are no restrictions on the amount of text you can include on your poster. However, ensure that the poster is clear, easy to read, and comprehensible when printed and displayed on a wall. The text should be legible to someone standing in front of the poster, allowing them to understand the content easily.

- **Encouraging Creativity:** Posters are visual mediums that allow for a high degree of creativity, which is exactly what we want to see from students. We encourage you to use a variety of mediums such as text, images, photos, and illustrations. Combining these elements creatively can help you convey your project's details effectively. The goal is to ensure that a reader can quickly understand what you have done and how you have done it.

- **Visual Content for Posters:** Students are encouraged to take photos of their own IoT systems and use them in their posters. Additionally, they can create illustrations by combining different icons and figures gathered from the Internet. It is essential to attribute the original sources of any images you use. Proper attribution not only respects the intellectual property rights of the creators but also enhances the credibility of your work. Ensure that all visual content is clear, relevant, and effectively contributes to the overall presentation of your project.

- **Demonstrating Critical Thinking:** Your poster should also demonstrate critical thinking and the ability to justify the decisions you made during the system design and

development process. Explain why you chose specific technologies, methodologies, and approaches and how these decisions contributed to the success of your IoT system. Highlight the rationale behind your design choices, addressing both the benefits and potential trade-offs. This will showcase your analytical skills and depth of understanding, providing a comprehensive view of your project's development journey.

- **Balancing Text and Visual Elements:** It is up to the students to determine the balance of text and visual elements in the poster. However, please note that the poster must be self-explanatory. Examiners should be able to read and look at the poster and understand its content without much effort and certainly without any assistance from the students. Strive for clarity and coherence, ensuring that both textual and visual elements complement each other to effectively convey your project's key points.

- **Importance of Clear and Concise Presentation:** Students are encouraged to research how best to organise, create, and visualise posters. There are various types of posters with different intentions. Some posters assume a presenter will be standing nearby to explain the project in further detail; therefore, the poster is used as an additional visual aid. However, in this case, the poster should be self-explanatory. Therefore, some advice provided in sources about poster creation needs to be considered carefully, as it may not serve the purpose of this project. Make your poster comprehensive and self-explanatory to ensure it stands alone effectively.

- While this project is not about creating the most beautiful posters, students should put significant effort into organising and presenting their content clearly and concisely. Readability is directly linked to the marks, as examiners need to be able to read and understand your material. A well-organised and easy-to-read poster will effectively communicate your project's details and contribute to a higher overall score.

- **Examples for Inspiration:** While we do not want to influence your creativity, the following posters contain a variety of elements that can be adopted during the poster creation process. None of the following posters are perfect or suitable for this project as they are. They are taken completely out of context to demonstrate some visual elements and show different ways of combining figures, text, illustrations, annotations, and labelling to create a creative, easy-to-understand, and informative poster. Please

do not try to copy any of the posters below as it is. Tthey are provided as starting points to explore how best to create a poster for your project.

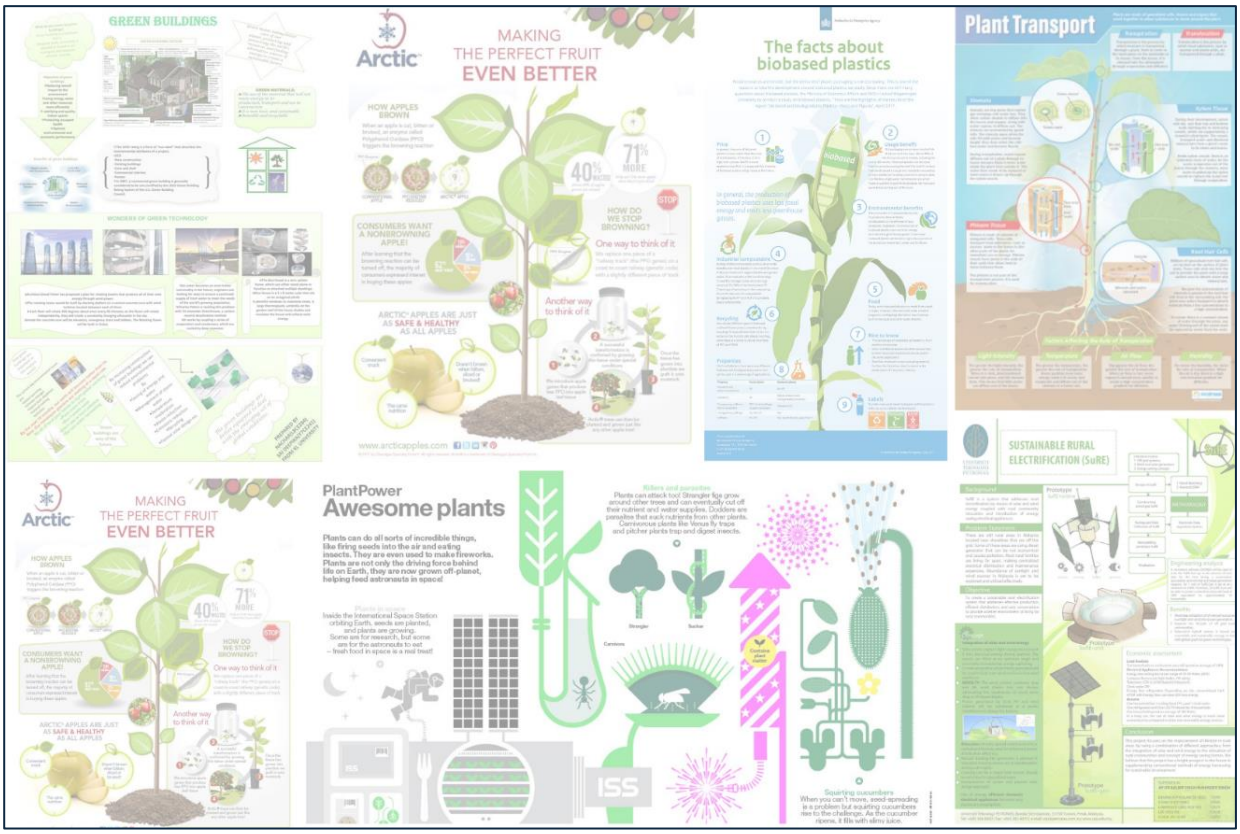


Figure 1: Sample Posters and Visual Elements

Students <u>must</u> label the poster with feature numbers. To position each feature on the poster, use a red circle (RGB: 192, 0, 0) with a 2 cm diameter and place the feature number inside the circle in white text. The short name of the feature and the critical discussion related to that feature should be located in a nearby area. This visual cue will help clearly identify and organise each feature, making it easier for examiners to follow your work. **The students will receive penalties (-20%) for not complying with this instruction.**

Demo Video (one video file in .mp4 format)

- Ideally, someone with minimum technical knowledge (e.g. future employer, charity officer, city council leader, etc.) should be able to understand your project by watching your demo video. This is an example: https://youtu.be/pa-ti02pHyk
- Avoid discussing or displaying code-level details in your demo video. Code-level explanations are reserved for CodeCast. Instead, concentrate on showcasing how the features you developed function in real-world scenarios.
- Besides demonstrating the features of your project, your demo video should also cover the following aspects:
    - **Problem Explanation:** Clearly describe the problem you addressed and highlight its importance and significance. Why is solving this problem crucial?
    - **Proposed Solution:** Explain the solution you proposed and provide details of what you built. What makes your solution effective?
    - **Potential Beneficiaries:** Identify and describe who the potential beneficiaries of your project are. Who will benefit from your solution?
    - **Overall Architecture:** Provide an overview of the overall architecture of your project. How are the different components of your project organised and interconnected?
    - **Feature Demonstration:** Briefly demonstrate all the functionalities of your project. Show how each feature works in a practical setting.
    - **Challenges Faced:** Discuss the main challenges you encountered during the development of your project. What obstacles did you have to overcome?
    - **Lessons Learned:** Share the lessons you learned throughout the project. What insights did you gain, and what would you do differently if you were to start again?
    - **Future Development:** Explain how you could further develop this project. What are the potential next steps or improvements?
- You are encouraged to edit your video to ensure it is clear, easy to understand, and engaging. Please make the content appealing and exciting for a wider audience.

CodeCast Video (one video file in .mp4 format)

- **Purpose of CodeCast**: Students should use the CodeCast video to explain their developed IoT system, focusing specifically on the code level. CodeCast is a screencast tool (https://en.wikipedia.org/wiki/Screencast) designed to explain code in detail.

- **Showing the Code**: You must display the actual code in the IDE environment you used for programming. Avoid using code snippets or screenshots.

- **Running the Code**: During the recording, run the code live to demonstrate that it is working. If the code is partially working or not working, still show it and provide reasoning for any issues.

- **Recording Techniques**: If screen-capturing software alone is insufficient to demonstrate that the code and system are working, feel free to use mobile devices to capture the screen or any other parts of the system. We encourage students to use a combination of mobile phone recordings and screen-capturing software.

- **Clarity and Verification**: Given that you might be using specific third-party frameworks, it may be difficult for examiners to replicate your environment and programming settings. Therefore, it is your responsibility to clearly show that your program is running live in the video. If examiners cannot determine whether the system is actually working, it could negatively affect your marks.

- **Editing the Video**: You are welcome to edit the CodeCast video to eliminate unnecessary segments. However, ensure that you do not remove any vital parts that help examiners understand and verify that your system is working.

- **Explaining the System**: Imagine you are sitting next to an examiner and explaining the system you developed. Clearly identify which parts of the system you adopted from the Internet and which parts you developed yourself. Explain this in both the written answers and the video. Students are encouraged to adopt source code and use frameworks from the Internet, but please do not claim them as your own work. Comment your code appropriately to indicate third-party code segments and explain

any modifications made.

- **Consistency**: Ensure that the answers provided in writing on the poster and the video are consistent. For example, if your written submission claims that you implemented a particular security feature, you need to show and explain this feature in the video to receive marks. Unsubstantiated claims in the written answers without corresponding evidence in the video will be ignored.

- **Claims and Evidence**: A common issue in marking complex projects is students making claims in their written submissions without actual implementation. Show the code details and explain them sufficiently in the video to help staff award the correct marks for the features you developed.

- **Marking Criteria**: To receive marks, you need to first claim the feature in the written submission (e.g., poster), then show how it works and explain the code in the video. Features shown in the video but not discussed in the written submission will be ignored. The CodeCast video serves as evidence, so ensure all claims in the written answers are backed up with detailed explanations in the video. Use feature numbers and their corresponding short names to link the discussion across all four submissions.

- **Code Repository**: All claims should be accurately reflected in your code repository. Although we primarily review the CodeCast video and written submission, we reserve the right to investigate the code repository further if discrepancies are suspected. If we find that the submitted code repository does not match the CodeCast or written submission, penalties may be applied. Therefore, ensure that your written submission, CodeCast video, and code repository reflect an identical IoT system.

Organising the Video: Students **must** organise the video in the order of feature numbers and discuss each feature one by one. The staff reserve the right to impose penalties (up to -20% marks) for students who violate this order of showing evidence in their video.

Source Code (one .zip file)

- **Accessibility:** Ensure that there are no passwords or other access restrictions on the .zip file. Anyone who downloads the file should be able to unzip and check the source code without any issues.

- **Source Code Management:** Throughout the project, students are encouraged to use source code management services such as GitLab, GitHub, or similar platforms. These services provide version control and collaborative features that can greatly aid in the development process.

- **Format:** At the end of the project, you must submit a single .zip file containing all the source code. This is mandatory, and no links to external repositories (e.g., GitHub, GitLab) will be accepted.

- You need to include a **.md (Markdown)** https://en.wikipedia.org/wiki/Markdown file named **README** in the root folder of the zip file, which explains how you have organised the code, the folder/file structures and any details of 3rd party software and framework you have been using. You need to include sufficient details that a 3rd party can read and understand and set up and replicate your IoT system.

- **Folder and File Structure:** A detailed description of the directory layout and the purpose of each folder and file. This helps others understand how the project is organized.

- **Setup Instructions:** Step-by-step instructions on how to set up the project. This should include:
    - ***Prerequisites:*** List any software, libraries, or frameworks that need to be installed before setting up the project.
    - ***Installation:*** Detailed commands and steps to install dependencies and set up the environment.
    - ***Configuration:*** Instructions on how to configure the project, including any environment variables or configuration files that need to be set.

- **Running the Project:** Explain how to run the project, including:
    - Commands to start the application.
    - Instructions to run any included scripts or tools.
    - Examples of expected output or behavior.

- **Third-Party Software and Frameworks**: Provide details of any third-party software, libraries, or frameworks used in the project. This includes:
    - Names and versions of the software/frameworks.
    - Purpose of each third-party component.
    - Links to their official documentation.

- **Code Documentation**: Mention any in-line code comments, docstrings, or additional documentation files that explain the code in more detail.

- **Troubleshooting**: Common issues that might arise during setup or usage and their solutions.

Use of Third-Party Software Tools and Frameworks

There are no restrictions on using software to develop your IoT system. However, students are encouraged to prioritise the use of open-source software as much as possible. The use of third-party software tools and frameworks must be properly acknowledged in all four submissions: the poster, demo video, CodeCast, and source code. This ensures transparency and gives proper credit to the developers of your tools, frameworks, and other sample source code you may be extending or incorporating into your IoT system.

Demonstrating a High Level of Competencies

To achieve higher marks in this assessment, students are required to go beyond the core topics taught in the module and demonstrate a higher level of technical competence. We have deliberately provided multiple opportunities for you to explore advanced tools, frameworks, and integrations that can extend the functionality of your IoT system and showcase your ability to learn independently.

- **Use of Camera Module (or use Phone Camera):** Although the module does not provide detailed instruction on integrating camera modules, students who choose to include them can exhibit advanced problem-solving and research skills. By consulting online documentation, experimenting with available APIs, and adapting the module to your specific IoT use case, you will demonstrate that you can learn new technologies independently—a critical competency in the IoT domain.
- **Incorporating Cisco Packet Tracer:** Another way to showcase a high level of expertise is by incorporating hybrid simulation tools such as Cisco Packet Tracer. For instance, you could simulate part of your system in the virtual environment and enable real-world communication to trigger physical actuations. An example scenario might involve a virtual sensor detecting a condition in Cisco Packet Tracer and then signalling a real-world alarm connected to your physical IoT kit. Successfully implementing such virtual–physical interplay indicates both creativity and a deep understanding of networking and systems integration.

We encourage you to use your preferred search engine to discover new frameworks, systems, or APIs that could augment your IoT project. For every integration you attempt, ensure it adds substantial complexity and value—merely plugging in a trivial function is less likely to demonstrate the range of competencies required for higher marks. **We aim to add more options here**: https://ngcharithperera.github.io/iot-integration-library/

Use of AI Tools to Support Coding

Students are encouraged to employ AI tools for code generation as they deem appropriate.

Use of IoT Cloud Platform

All IoT projects should utilise some form of IoT cloud platform. Our default recommendation is ThingsBoard. You will receive free access to a university-hosted ThingsBoard account, which allows you to customise, configure, and program as needed. However, students are free to use any IoT cloud platform of their choice. In such cases, the students must ensure they have sufficient access to the chosen IoT cloud platform to complete the project and that the platform will be available throughout the semester. Students may also use open-source IoT cloud platforms, but they are responsible for hosting their own IoT cloud in these circumstances. Make sure to verify the platform's capabilities and availability to avoid any disruptions in your project workflow.

Access to IoT Cloud Platform

Each project group receives a unique username and password allowing access to the iot cloud platform (ThingsBoard). Hold the username and password listed in 'Access to IoT Cloud Platform (ThingsBoard)' inside the Technical Services folder in the Learning Central. Please make sure to change the password as soon as possible.

Web Hosting

Students can host their websites or simple web applications on either the OpenShift platform or the Users Web Server.

- https://cf.sharepoint.com/sites/COMSCDocumentation/SitePages/Using-Open-Shift.aspx
- https://cf.sharepoint.com/sites/COMSCDocumentation/SitePages/Creating-and-Publishing-a-Website-on-the-School's-Users-Web-Server.aspx

If you encounter any issues or require further assistance, please contact IT Support at IT-Support@cardiff.ac.uk

Use of IoT Kit/Hardware

Each group will have access to an IoT hardware kit (i.e., one identical to the ones you use during labs). Refer to the sheet in the IoT kit bag to find the full IoT components available for your coursework. You should only use the IoT kit provided by the school for your coursework. You are allowed to connect your mobile phones and laptops to the IoT system. **No additional electronic components or devices of any kind are allowed.**

Access to IoT Kit Hardware

- Check the Technical Service Notice Board within the Technical Service Folder in Learning Central to understand how to collect your IoT kit (or via the Link).The Technical Services team will routinely publish updates on the notice board to keep you updated. Check the noticeboard before emailing them.
- Contact comsc-technical@cardiff.ac.uk for any problem with IoT Kits (e.g., Not working, lost, malfunctioning, etc). Please make sure to mention that you are contacting them regarding the CMT223 module. Otherwise, the technical team might not be able to assist you efficiently.

Extenuation Circumstances

Please follow the school-recommended standard Extenuating Circumstances application process. Please do not contact the module leader of this module asking for Extenuating Circumstances extensions, as individual module leaders do not have the authority to approve any Extenuating Circumstances or extensions.

Deal with Group Problems and Individual Marks

- BuddyCheck will be used to convert group marks into individual marks by considering the contributions of the individual students.
- Students are expected to get familiar with how BuddyCheck works using this tutorial: https://xerte.cardiff.ac.uk/play_20049
- BuddyCheck's decision will be final and cannot be challenged.

- Participation in BuddyCheck is compulsory, and students who do not participate in BuddyCheck may be disadvantaged compared to those who do.
- BuddyCheck will request feedback from students four times during the semester. The first three times aimed to motivate students and encourage positive group engagements. Students will receive peer feedback from BuddyCheck about their contribution.
- BuddyCheck will use the Fourth (last) round of feedback to adjust the group mark to an individual mark by multiplying the group mark with the BuddyCheck generated weighting.

Frequently Asked Questions (FAQ)

All students must regularly review the Frequently Asked Questions section on Learning Central each week and before posing any inquiries or sending emails, as their questions might already be addressed there.

Assessment Criteria

1. Problem Identification and Project Overview (5%)

High Distinction 80%+	Clearly identifies a significant problem with original insights. Provides a thorough, sophisticated explanation of the project overview and use cases. Demonstrates excellent understanding of the context and existing solutions. All details are well-structured, no grammatical errors, and exceptional presentation.
Distinction 70-79%	Identifies a significant problem with detailed explanation. Comprehensive project overview and use cases. Strong understanding of the context and existing solutions. Well-structured, minimal grammatical errors, and excellent presentation.
Merit 60-69%	Identifies a relevant problem with clear explanation. Detailed project overview and use cases. Good understanding of the context and existing solutions. Well-structured, minor grammatical errors, and good presentation.
Pass 50-59%	Identifies a basic problem with adequate explanation. Adequate project overview and use cases. Basic understanding of the context and existing solutions. Adequately structured, some grammatical errors, and clear presentation.
Marginal Fail 40-49%	Identifies a problem with limited explanation. Basic project overview and use cases. Limited understanding of the context and existing solutions. Poorly structured, multiple grammatical errors, and unclear presentation.
Fail 0-39%	Poorly identifies a problem with minimal explanation. Incomplete project overview and use cases. Minimal understanding of the context and existing solutions. Very poorly structured, numerous grammatical errors, and very unclear presentation.

2. Architecture and System Design (5%)

High Distinction 80%+	Exceptionally detailed and innovative system architecture. Highly effective and justified design choices. Excellent use of diagrams and annotated figures. Perfect structure, no grammatical errors, and excellent presentation.
Distinction 70-79%	Detailed and innovative system architecture. Well-justified design choices. Effective use of diagrams and annotated figures. Well-structured, minimal grammatical errors, and excellent presentation.
Merit 60-69%	Clear and detailed system architecture. Justified design choices. Good use of diagrams and annotated figures. Well-structured, minor grammatical errors, and good presentation.
Pass 50-59%	Adequate system architecture with some details. Some justification for design choices. Basic use of diagrams and annotated figures. Adequately structured, some grammatical errors, and clear presentation.
Marginal Fail 40-49%	Basic system architecture with limited details. Limited justification for design choices. Poor use of diagrams and annotated figures. Poorly structured, multiple grammatical errors, and unclear presentation.
Fail 0-39%	Poor system architecture with minimal details. Minimal justification for design choices. Very poor use of diagrams and annotated figures. Very poorly structured, numerous grammatical errors, and very unclear presentation.

3. **Implementation and Functionality (Sensing and Actuation, Networking and Communications, Data Management and Analytics) (30%)**

High Distinction 80%+	Exceptionally functional and innovative implementation. All components (sensing, actuation, networking, data management) are integrated flawlessly. Demonstrates advanced understanding and application of IoT principles. No flaws in functionality, highly effective solution.
Distinction 70-79%	Highly functional and innovative implementation. All components are well integrated. Demonstrates strong understanding and application of IoT principles. Minor flaws in functionality, very effective solution.
Merit 60-69%	Functional and innovative implementation. Most components are well integrated. Demonstrates good understanding and application of IoT principles. Some minor flaws in functionality, effective solution.
Pass 50-59%	Adequate implementation with some innovation. Some components are integrated. Demonstrates basic understanding and application of IoT principles. Some noticeable flaws in functionality, somewhat effective solution.
Marginal Fail 40-49%	Basic implementation with limited innovation. Limited integration of components. Demonstrates limited understanding and application of IoT principles. Significant flaws in functionality, less effective solution.
Fail 0-39%	Poor implementation with minimal innovation. Minimal integration of components. Demonstrates minimal understanding and application of IoT principles. Major flaws in functionality, ineffective solution.

4. Privacy and Security (10%)

High Distinction 80%+	Exceptional implementation of privacy and security measures. Highly innovative and well-justified approaches. Demonstrates advanced understanding of privacy and security principles. No flaws in privacy and security implementation.
Distinction 70-79%	Highly effective implementation of privacy and security measures. Innovative and well-justified approaches. Demonstrates strong understanding of privacy and security principles. Minor flaws in privacy and security implementation.
Merit 60-69%	Effective implementation of privacy and security measures. Well-justified approaches. Demonstrates good understanding of privacy and security principles. Some minor flaws in privacy and security implementation.
Pass 50-59%	Adequate implementation of privacy and security measures. Some justification for approaches. Demonstrates basic understanding of privacy and security principles. Some noticeable flaws in privacy and security implementation.
Marginal Fail 40-49%	Basic implementation of privacy and security measures. Limited justification for approaches. Demonstrates limited understanding of privacy and security principles. Significant flaws in privacy and security implementation.
Fail 0-39%	Poor implementation of privacy and security measures. Minimal justification for approaches. Demonstrates minimal understanding of privacy and security principles. Major flaws in privacy and security implementation.

5. Human Factors and Interaction (10%)

High Distinction 80%+	Exceptionally user-centered design and interaction. Highly innovative and intuitive interfaces. Demonstrates advanced understanding of human factors. No flaws in user interaction and experience.
Distinction 70-79%	Highly effective user-centered design and interaction. Innovative and intuitive interfaces. Demonstrates strong understanding of human factors. Minor flaws in user interaction and experience.
Merit 60-69%	Effective user-centered design and interaction. Intuitive interfaces. Demonstrates good understanding of human factors. Some minor flaws in user interaction and experience.
Pass 50-59%	Adequate user-centered design and interaction. Somewhat intuitive interfaces. Demonstrates basic understanding of human factors. Some noticeable flaws in user interaction and experience.
Marginal Fail 40-49%	Basic user-centered design and interaction. Limited intuitive interfaces. Demonstrates limited understanding of human factors. Significant flaws in user interaction and experience.
Fail 0-39%	Poor user-centered design and interaction. Minimal intuitive interfaces. Demonstrates minimal understanding of human factors. Major flaws in user interaction and experience.

6. Design Strategies and Prototyping (8%)

High Distinction 80%+	Exceptionally innovative design strategies and prototyping. Highly effective and well-justified approaches. Demonstrates advanced understanding of design principles. No flaws in design strategies and prototyping.
Distinction 70-79%	Highly effective design strategies and prototyping. Innovative and well-justified approaches. Demonstrates strong understanding of design principles. Minor flaws in design strategies and prototyping.
Merit 60-69%	Effective design strategies and prototyping. Well-justified approaches. Demonstrates good understanding of design principles. Some minor flaws in design strategies and prototyping.
Pass 50-59%	Adequate design strategies and prototyping. Some justification for approaches. Demonstrates basic understanding of design principles. Some noticeable flaws in design strategies and prototyping.
Marginal Fail 40-49%	Basic design strategies and prototyping. Limited justification for approaches. Demonstrates limited understanding of design principles. Significant flaws in design strategies and prototyping.
Fail 0-39%	Poor design strategies and prototyping. Minimal justification for approaches. Demonstrates minimal understanding of design principles. Major flaws in design strategies and prototyping.

7. Poster Quality (8%)

High Distinction 80%+	Exceptionally clear and visually engaging poster. Innovative use of design elements to convey information effectively. Thorough and insightful explanation of all aspects of the project. Perfect structure, no grammatical errors, and excellent presentation. Exceptional balance of text and visuals; highly professional.
Distinction 70-79%	Comprehensive and well-designed poster. High-quality visuals and clear text. Thorough explanation of project aspects. Well-organized and easy to follow. Excellent balance of text and visuals; very professional.
Merit 60-69%	Clear and well-structured poster. Good use of visuals and text to explain the project. Detailed explanation of key project aspects. Well-organized and mostly easy to follow. Good balance of text and visuals; professional.
Pass 50-59%	Adequate poster with clear explanations. Basic use of visuals and text; minor issues with clarity. Explanation of project aspects is present but may lack depth. Organized but may have some readability issues. Adequate balance of text and visuals.
Marginal Fail 40-49%	Basic poster with limited clarity. Limited use of visuals and text; significant issues with clarity. Basic explanation of project aspects; lacks depth. Organization is present but may be confusing. Limited balance of text and visuals.
Fail 0-39%	Poorly designed poster with major clarity issues. Minimal use of visuals and text; very difficult to follow. Incomplete or unclear explanation of project aspects. Poor organization and readability. Imbalance of text and visuals; unprofessional.

8. Demo Quality (8%)

High Distinction 80%+	Exceptionally clear and engaging demonstration. Innovative presentation techniques. Thorough and insightful explanation of all aspects. Perfect video and audio quality. Excellent clarity, production quality, and engagement.
Distinction 70-79%	Comprehensive and engaging video demonstration. High video and audio quality with clear visuals. Thorough explanation of problem, solution, and beneficiaries. Well-planned and scripted; high production quality.
Merit 60-69%	Well-produced video with clear and concise demonstration. Good video and audio quality. Effective explanation of problem, solution, and beneficiaries. Detailed overview of architecture and feature functionality.
Pass 50-59%	Clear demonstration of features and functionality. Adequate video and audio quality; minor issues present. Basic explanation of the problem, solution, and beneficiaries.
Marginal Fail 40-49%	Basic demonstration of features with significant flaws. Limited clarity in video and audio. Some explanation of the problem and solution but lacks depth.
Fail 0-39%	Limited demonstration of features; major flaws in clarity. Poor video and audio quality. Incomplete or unclear explanation of problem and solution.

9. CodeCast Quality (8%)

High Distinction 80%+	Exceptionally clear and engaging code explanation. Highly detailed and insightful. Perfect alignment with written submission. Excellent video quality and clarity. Outstanding code explanation and demonstration.
Distinction 70-79%	Comprehensive and insightful code explanation. Detailed and clear demonstration of functionality. Thorough alignment with written submission. High video quality.
Merit 60-69%	Well-explained and detailed code walkthrough. Clear demonstration of functionality. Consistent and well-aligned with written submission. Good video quality.
Pass 50-59%	Clear explanation of code with minor issues. Adequate detail and clarity in code demonstration. Consistent with written submission.
Marginal Fail 40-49%	Basic explanation of code with significant flaws. Limited detail and clarity in code demonstration. Inconsistent with written submission.
Fail 0-39%	Limited explanation of code; major flaws in detail. Poor video quality and unclear code demonstration. Significant inconsistencies with written submission.

10. Source Code Quality and Documentation (8%)

High Distinction 80%+	Exemplary organization and readability of code. Thorough and insightful documentation and comments. Outstanding code quality and functionality. README file is highly detailed and perfectly aligns with CodeCast video. Exceptional clarity and completeness in code documentation.
Distinction 70-79%	Exceptionally organized and readable code. Comprehensive documentation and comments. High-quality and functional code. README file is thorough and aligns with CodeCast video.
Merit 60-69%	Well-organized and readable code. Good documentation and comments. Functional and well-structured code. README file is detailed and clear.
Pass 50-59%	Clear organization and readability of code. Adequate documentation and comments. Functional code with minor issues. README file provides basic explanations.
Marginal Fail 40-49%	Basic organization of code; limited readability. Minimal documentation and comments. Functional but with significant issues and errors. README file is present but lacks detail.
Fail 0-39%	Poorly organized code; minimal readability. Inadequate documentation and comments. Partially functional code with major issues. Incomplete README file.

Help and Support

- Questions about the assessment can be asked in person in weeks 1 to 12 of the Spring Semester during office hours (the same as your lab sessions) or via email.
- My office hours correspond with the scheduled Lab sessions (week 1 to 6). Each week, you'll have a 2-hour slot for lab work. Generally, labs take between 45 to 90 minutes, varying based on the group's skill level. I encourage you to communicate with me during those times. Please contact me in advance to schedule a meeting to prevent any issues, or feel free to visit on the day and I will accommodate you on a first-come, first-served basis.
- This module utilises email and in-person interactions only for streamlined and productive communication. Kindly refrain from using Microsoft Teams, Discord, or other platforms for inquiries.
- Responses may be anonymised and broadcast via Learning Central Announcements if the response to the question would benefit other students.
- I aim to respond to questions within 3 working days. You are encouraged to send a follow-up email if you do not receive a reply within 3 working days.

Feedback

Feedback on your coursework will address the above assessment criteria. Feedback and marks will be returned via Learning Central in accordance with the university's Academic Feedback Policy. You can see feedback return dates for each assessment on Learning Central: see **'COMSC Assessment Calendar 2025 - 26'** page.